



5 (a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 5 \cos x. \quad [7]$$

[illegible]



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- (b) For large positive values of x and for any initial conditions, show that the solution to part (a) can be approximated by

$$y \approx R \sin(x + \phi),$$

where the constants R and ϕ are to be determined. [3]

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